

Topic Proposal



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Guitar Tuner System



The Problem

Many beginner guitar players experience difficulty tuning their instruments accurately. Traditional tuning methods rely on a person's hearing ability, which may lead to incorrect tuning. Without proper tuning tools, users may struggle to identify whether the sound of a guitar string is too low or too high.



Proposed Solution

Develop a C++ Based Guitar Tuner System, an application that detects the sound frequency of guitar strings and determines whether the string is correctly tuned.

The system will:

- Detect sound frequency using the microphone.
- Compare detected frequency with standard guitar tuning.
- Identify the closest guitar string.
- Display whether the string is Low, High, or In Tune.



Features/Description

- Start Tuner – Begins the guitar tuning process
- Capture Audio – Records sound from the microphone
- Detect Frequency – Analyzes the pitch of the sound signal
- Identify String – Compares detected frequency with standard tuning
- Display Result – Shows the detected string and tuning status



Data Structure

The system utilizes an array data structure to store the standard frequencies and corresponding names of the six guitar strings, allowing efficient comparison between the detected frequency and the reference values. A structure (struct) is also used to organize each string's data, grouping the string name, standard frequency, and tuning status together as a single record for easier access and processing.



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Thank You!!